CPG · MANUFACTURING PLANNING

CPG · Product requirements

A practical MVP supply-chain planning and execution system for a small manufacturer. Requirements, decisions, operating rules, and a visual domain model in one readable source.

MVP functional baseline | 19 October 2026 | English source of truth

No stack or deployment decisions

01 / CPG: product and scope

CPG is a deliberately minimal, genuinely usable planning and execution tool for a small manufacturer. The name is simply CPG; *Caveman Planning Good!* is its playful footnote, not a second name.

2 roles	2 tenants	90 days
One built-in admin plus normal users	Test and Prod, isolated	Daily planning, weekly display aggregation

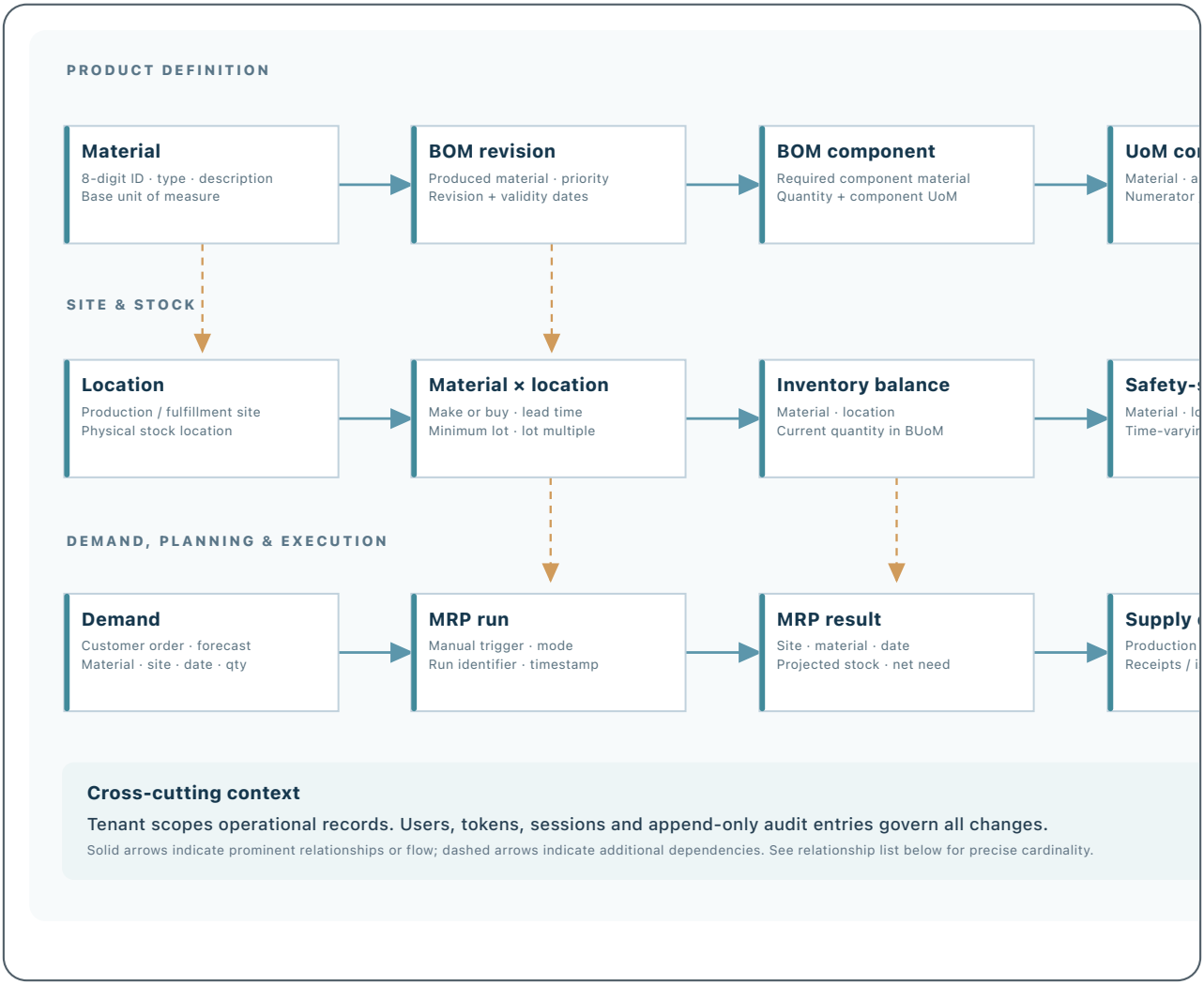
In scope: materials, UoM, immutable/alternative BOMs, locations, customers, suppliers, sales orders and forecasts, safety stock, MRP, production, purchasing, stock ledger/corrections/transfers, reports, audit, XLS imports, full API parity, two languages, light/dark modes, onboarding documentation and contextual tooltips.

Out of scope: finance, costing, invoices, refunds, a customer-return process, stock batches/serial numbers, machine/work-center/capacity scheduling, hard reservations, working-day calendars, SSO, approval workflows, multiple administrators, supplier price/source lists, automatic stock-transfer creation, or any particular technology stack, hosting and deployment approach.

> **Requirements baseline:** decisions explicitly agreed by the product owner and small closure defaults have been consolidated. This specification plus the visual conceptual model and decision register constitute the MVP baseline. New modules must not be inferred from enterprise ERP terminology.

02 / Conceptual data model

The model ties product definitions to site-specific stock and requirements. The visual below is intentionally conceptual rather than a physical database schema.



Conceptual overview, not one database table per box. See visual conceptual data model for BOM versions, multi-line documents, in-transit stock and audit relationships.

> **Immutable BOMs:** Any actual change to a logical BOM's recipe, components, priority or validity creates a new immutable revision. Activation/deactivation changes only the logical BOM's audited active flag and does *not* generate a new recipe revision. Identical imports do not. Production orders pin their original revision even when superseded or deactivated. Equal priorities for different *currently selectable* BOM alternatives producing the same material are forbidden when validity periods overlap, even partially. Validate on any edit. Reject direct/indirect BOM cycles with the cycle shown.

Materials & UoM

One tenant-local eight-digit zero-padded sequential material ID (starting `00000001`, never reused within one dataset generation; a Test

BOM structure

A recipe specifies raw/intermediate components required for **one BUoM of the output**, scaled proportionally. Only intermediate or finished

Within one dataset generation, a reset starts a new generation and restarts numbering), required description, type (raw/intermediate/finished) and base UoM. Convert immediately from entered UoM to BUoM; retain entered unit and factor in history. Store authoritative BUoM quantities to six decimal places using decimal, never binary floating-point, arithmetic. User-entered alternate-unit conversions round half-up to six BUoM decimals once at entry; MRP components and supply round upward. Preserve entered quantities/factors. Conversion edits never revalue existing BOMs or transactions.

Only intermediate or finished materials can have BOMs. Raw materials cannot be BOM outputs, and finished materials cannot be BOM components. Material type cannot change after BOM or transaction reference.

Locations and planning

Physical sites hold inventory. At each material/location define Buy/Make, whole-day lead time, minimum quantity and lot-size multiple. Raw is Buy-only; intermediate/finished can be either. Safety stock is effective-dated, valid until the next record and zero if undefined.

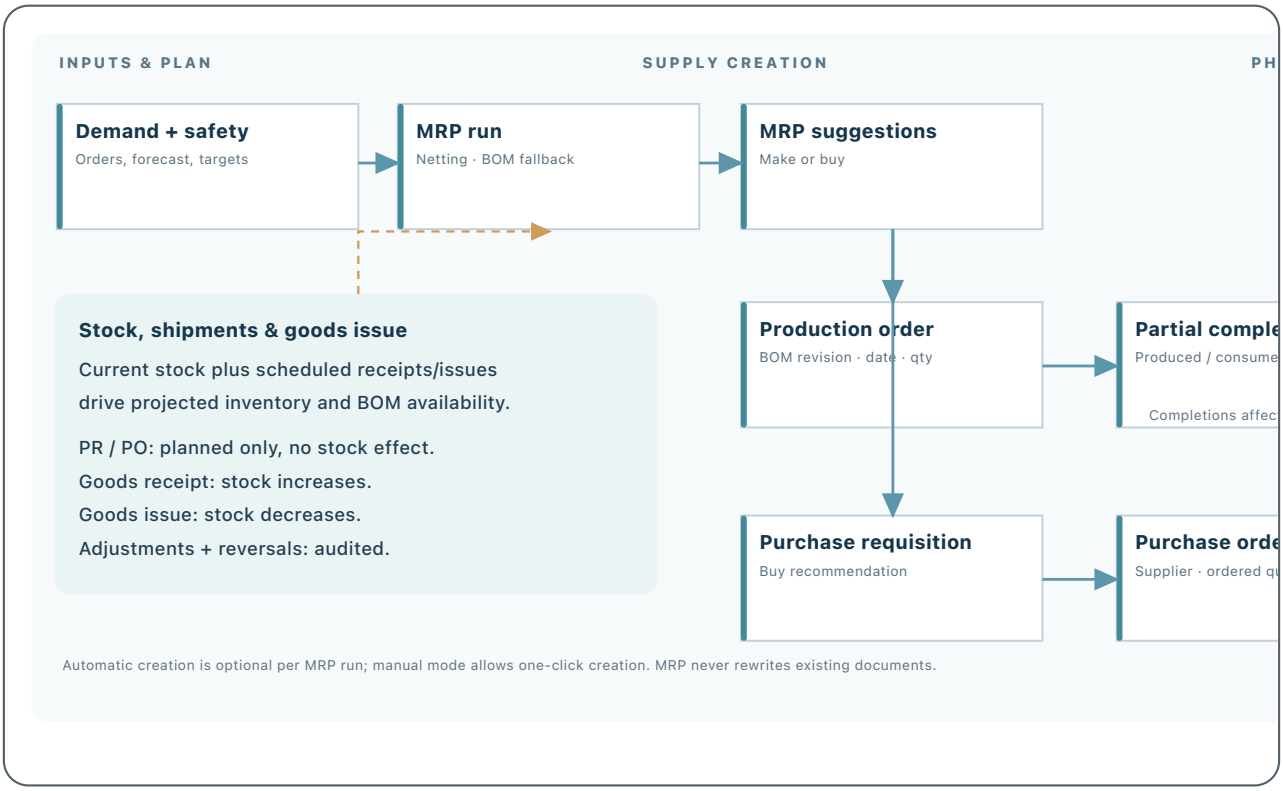
Masters and integrity

Customers and suppliers each have ID/name. Inactive material/location cannot strand open stock or activity. Business IDs and records are strictly tenant-local; no cross-tenant references. BUoM locks when a material is first referenced by *any* BOM revision, operational document or inventory movement. An active BOM reference alone does not block material deactivation, but its inactive

components make the recipe ineligible for new MRP orders; use an eligible alternative or raise an explicit parent exception. Conversion

changes apply only to new entries.

03 / End-to-end operational flow



Conceptual flow, not actual warehouse transactions. Physical stock changes on postings only.

Operational behaviors

Area	Required MVP behavior
Production orders	One output material per order, one location and a pinned BOM revision. Planned completion is order date; planned start = completion minus whole-calendar-day lead time. Manual users may choose another date-valid BOM even when material is unavailable, with a warning; completion still checks stock.
Production postings	Partial output and consumption postings, including **zero output with consumed components** as waste, are allowed. CPG pre-fills expected inputs for produced quantity from BOM, but actual consumption may differ or include extra raw/intermediate material. Variances display; overproduction gives warning and is permitted. Complete order and stock changes are atomic.
Procurement	PR (internal request) → PO (supplier selected) → GR (physical acceptance),

chain	but users may also create a PO directly. One PR can be converted into several POs, potentially with different suppliers. Any active supplier may supply any material. Supplier-specific costs and lead times are excluded.
Multi-line document model	Customer orders and POs have one header-level location and multiple lines, each with material, UoM, quantity and date. Production orders and transfers remain single-material. One goods receipt covers one PO and may contain multiple PO lines; one outgoing shipment covers one customer order and may contain multiple order lines. No mixing of several parent orders into one receipt/shipment.
Receipt and issue	Actual GR increases physical stock; PO receipt may be partial or in excess of ordered quantities (warn, not block). Actual GI decreases stock and fulfills customer-order lines; outbound quantities must never exceed outstanding customer-order line amounts. Planned shipments themselves do not change stock.
Operational edits	After customer GI, fulfilled line's material/location/original due date locked; ordered quantity may not fall below delivered. After PO GR, supplier/location locked; received line material/UoM locked; quantity may not fall below received; dates editable. After first production completion, material/location/BOM revision locked; target cannot fall below output already posted; planned finish can change. All edits audited.
Stock movement and correction	All movements go into a signed BUoM ledger separate from generic audit. Both user roles can post reasoned corrections, including opening balance. Excel import for opening stock and later inventory corrections contains desired *physical current stock*, not adjustment deltas. System calculates changes, validates dry run and posts atomically; absent rows remain untouched. Ordinary corrections never alter in-transit inventory.
Reversal transaction	GR and GI line postings, inventory adjustments and individual production completion postings can be reversed in full, at most once. Current business date, mandatory reason, original-posting reference and full audit. Other lines remain unchanged; partial error correction = reverse the entire line and post a corrected one. Completion reversal removes its produced stock and restores exactly actually consumed components; reject if produced stock is insufficient. Status/outstanding quantities recalculate. **Rare manual workaround:** when downstream use leaves insufficient stock for full reversal, a reasoned temporary positive inventory adjustment may make reversal possible. Complete the full reversal and corrected posting, then reverse the temporary adjustment too. Every intermediate and final balance must be nonnegative; competing stock writes should be avoided. Do not leave phantom stock.
Transfers	One material, different source/destination locations, planned dispatch and arrival (arrival never before dispatch). Partial issues create in-transit stock;

	partial receipts consume in-transit and raise destination stock (never receive more than dispatched). Transfers participate in MRP but are never auto-created. To Close Early, transit must be completely received or written off with reason. **Completed** requires 100% planned quantity physically received and zero write-off; any abandonment or write-off resolves as **Closed Early** once transit is zero. Reverse destination receipts and write-offs before source GI; reversal can automatically recalculate a previously closed transfer to Open.
Date and integrity	All actual GR, GI, production, adjustment, transfer, write-off and reversal postings use **today's business date**; no back- or forward-dated physical postings. Planning/document dates may be historical or future. All inventory movements are atomic and **physical or in-transit stock never becomes negative**. Customer returns: ordinary positive stock correction only; no return process.

Four-state lifecycle

Area	Required MVP behavior
Open	Active, including partially fulfilled; per-line progress recorded by quantities, not an extra status.
Completed	All lines fulfilled, received, converted or manufactured; for POs and production orders recorded amount may exceed planned.
Closed Early	Remaining active amounts intentionally abandoned after partial work. Physical history stays; MRP stops counting remaining unprocessed amounts.
Cancelled	Allowed only before any physical posting/conversion; keep document and audit. No manual reopening. Reversals may recompute prior status back to Open.

Order aggregation: each customer/PO line tracks its own fulfilled and closed quantities. Header is Open if any line remains active, Completed if all fulfilled, Closed Early if all resolved with at least one partially abandoned, and Cancelled only when nothing has been processed. A requisition is completed by conversion to PO, a PO by GR, a customer order by GI, a production order by completion. No manual reopen; a reversal can automatically recalculate status. Outstanding closed/cancelled quantities do not count in MRP.

Transfers are special: a transfer cannot be Completed or Closed Early while in-transit stock remains. All planned units received without loss → Completed; any written-off or abandoned amount → Closed Early once transit is zero. In-transit is not available as physical inventory at either site. Dedicated audited transfer write-off/reversal handles

discrepancies; general inventory correction does not.

04 / MRP: all agreed calculation rules

> **Invocation:** Manual on-demand runs only, for **all locations in the selected tenant**, over **90 daily buckets starting today**. Separate beyond-horizon recommendations may have later dates, without additional stock buckets. Results have run ID/time and remain visible until superseded; users may filter/aggregate results after running. Read-only tenant blocks new runs.

Area	Required MVP behavior
Demand and forecast consumption	By material/location/original due date, customer orders consume forecast: 100 forecast and 60 ordered = 60 firm + 40 residual. Outstanding need = **unfulfilled active order quantity + unconsumed forecast**. Goods-issued quantities reduce outstanding firm demand without restoring previously consumed forecast. Early closure/cancellation releases only the abandoned portion into residual forecast (e.g. 60 ordered, 30 fulfilled/30 abandoned → 70 residual forecast). **Past-day rule:** residual forecast dated before today's business date expires and never rolls forward; overdue unfulfilled firm-order demand instead moves to today, retaining original-date forecast consumption for history.
Overdue demand and priority	Unfulfilled orders originally due before today remain outstanding and need supply today, but still consume forecast on their original dates. Expired historical forecast residual never becomes catch-up demand. Priority: customer orders by earliest *original* due date, then current/future residual forecast, then safety target gap; orders on the same original date have equal business priority.
Projection and safety stock	Physical stock never negative; MRP may show negative projected balance with deficit and projected balance after recommendations. Safety stock effective from its date until changed, default zero; replenish to at least target applying minimum quantities and lot-size multiples.
Supply pipeline and overdue	Count remaining active PR/PO/production quantities and scheduled receipts/issues, avoiding double count of posted output and converted PR→PO. PR is *tentative*: prevents duplicate PR but not

	dependable for BOM feasibility; PO and valid executable production orders are dependable planned pipeline. **Exception:** if an existing production order's pinned recipe contains a subsequently deactivated component, retain the order unchanged but mark its unproduced balance **At-Risk Existing Production / Inactive BOM Component**. Show its planned commitment separately, exclude its output from dependable inventory/parent-BOM feasibility, and explicitly report blocked demand. Do *not* auto-create replacement output merely because of this at-risk order; user must resolve/reactivate component or close/cancel the existing order and rerun. Overdue eligible outstanding supply projects today with original dates/warnings unchanged.
BOM choice and explanation	Among active date-valid BOM revisions, calculate final output quantity **after** upward precision/minimum/lot-size rounding, then evaluate the full rounded component requirement against projected on-hand plus dependable already-scheduled receipts/issues (not new suggestions or tentative PR stock). Skip recipes containing inactive components for *new* planning, with explicit **Inactive BOM Component** explanation; try the next eligible alternative. Choose highest-priority feasible recipe, explain rejected higher-priority options and exact shortages. If none is eligible, raise a parent planning exception; no executable parent order.
Multilevel explosion	Recursively derive raw/intermediate requirements, rejecting cycles. When a parent BOM is not yet feasible, show the complete **provisional multilevel dependency plan** using the highest-priority date-valid active recipe with eligible active inputs: hypothetical parent and intermediate actions, component quantities, dates and blockers. This is a read-only scenario, *not* dependable projected supply or an executable recommendation. Do not assume its output to make another action feasible in the same run, net it as real supply or use it to suppress shortages. Independently feasible lower-level Buy PRs or Make orders remain executable. **Never auto-create infeasible parent orders**; rerun after dependable supply materializes.
Component timing and execution	Production planned start is completion minus location lead time; BOM inputs are reserved *in MRP projections only* on planned start day, while planned finished output arrives completion day. No hard physical reservations: actual inputs/output are posted at production completion. As production is partially completed, only the unproduced balance's BOM inputs remain planned (without extrapolating actual variance); overdue start-day component requirements shift to today.
Overdue and late starts	Production/PO output overdue is expected today with warning. If a proposed make/buy action's lead time implies starting in the past,

	plan earliest start today and feasible finish, flag Late Supply and quantify lateness without changing original demand date. The stock projection retains exactly 90 daily buckets today through day 89; recommendations whose earliest feasible finish lies later remain visible separately with **Beyond Horizon** dates/warnings and do not increase stock inside the 90 days. Same-day expected receipts can offset same-day projected demand but physical issues require physically posted stock.
Transfers	Plan expected dispatch at source and receipt at destination separately; in-transit not consumable. Only project receiving what the source can supply. Report Insufficient Transfer Stock, without automatically modifying or creating transfers. Overdue in-transit receipt expected today in projection with Overdue Transfer warning.
Parameters/precision	Raw = Buy only; intermediate/finished can Make or Buy by site. Missing site parameters or usable BOM → distinct MRP exception, without guessing or blocking unrelated materials. Decimal, never binary float, intermediates; authoritative BUoM quantities have six decimal places. Explicit alternate-UoM input uses its exact positive conversion factor, rounds *half-up* once into stored six-place BUoM, preserves original unit/factor and rejects invalid or rounded-zero inputs. Aggregate computed requirements precisely where possible, then round planned component requirements and new supply *upward* to six places. Apply output minimum/lot multiple *before* checking the resulting full BOM component feasibility; round those component requirements upward too. Inverse-UoM displays are indicative only and never revalue history.
Reruns and duplicate avoidance	Recalculate results/recommendations from scratch, retaining all PR/PO/production orders. Net their remaining active quantities and dependable receipts without double-counting; hypothetical proposals never become real supply. Existing at-risk production orders remain committed/visible but their output is excluded from dependable feasibility: raise explicit blocker and do not auto-create a competing replacement until their state is resolved. On falling demand show excess, never auto-cancel/modify real documents.
Manual/auto creation	Default user-confirmed one-click creates PR for Buy or production order for **lot-adjusted and fully feasible** Make. Explicit automatic mode at run start creates executable eligible recommendations only, all-or-nothing; provisional/blocked multilevel parent plans cannot be accepted or auto-created. On failure retain calculations/exceptions without new documents. Tag created documents with run ID and never modify existing transactions.
Stale and concurrent	Check present demand/stock/documents and relevant version

runs	numbers immediately before accepting a recommendation or auto-creating orders; if stale, block and require a fresh run. Only one automatic MRP creation operation per tenant at once. UI/API optimistic locking, atomicity and idempotency protect retries.

> **Closure defaults:** at daily granularity, same-day expected receipts may offset same-day requirements in projection; physical stock must exist before issue. Expired past forecasts never roll forward; round output lots before BOM feasibility and keep hypothetical multilevel plans separate. A tentative PR can suppress duplicate requisitions but is not reliable stock for a BOM feasibility test. Equal-priority customer orders on the same date remain equal; an internal stable-ID tie-breaker only keeps results deterministic.

05 / Users, tenants, imports, help and branding

Area	Required MVP behavior
One administrator	Exactly one built-in administrator. Password from mandatory runtime environment variable; app refuses to start if missing. No extra admins and no password change in the UI. Admin creates, deactivates and resets normal users; cannot delete them.
Roles and tenant access	Every active account can select Test or Prod; current tenant is clearly visible. Both roles maintain transactions, post inventory corrections, use reports and audit, run MRP when writable and manage own tokens. Only admin maintains master data, user accounts, read-only flags and Test reset.
Sessions and passwords	No SSO. Normal-user passwords set/reset by admin; users cannot change them. Unlimited concurrent sessions and shared account usage permitted. User deactivation or password reset immediately invalidates all sessions and API tokens of that user.
Strict tenant isolation	Test and Prod do not share business master data, transactions, inventory, MRP results, audit or ID sequences. Test is preloaded with demonstration records; Prod begins entirely empty. Global users, sessions, tokens, language/theme settings and Global audit survive a Test reset.
Tenant read-only control	Each tenant independently has an admin-set read-only flag. UI and API, including admins, cannot change tenant business data or run MRP while

	read-only. Data, results and audit can still be viewed. Login, personal preferences/tokens, admin user management, changing flag and protected Test reset remain available.
Atomic Test reset	Reset temporarily forces Test read-only, waits for or blocks in-flight writes, regenerates data as one atomic operation and wipes only Test business history and Test audit. **Start from scratch:** create a fresh Test dataset generation, restarting *all* Test-local business ID sequences from 00000001. Old Test IDs can reappear in a later generation but are never treated as references to old records. Invalidate prior Test-generation dry runs, MRP proposals and idempotency records. Test-scoped API mutations must carry the current dataset-generation precondition obtained from tenant metadata; the UI supplies it automatically. Old/missing generation is rejected (HTTP 409/428), so a late replay can never mutate a new object sharing an ID. Global users/tokens/preferences and Global audit survive. Restore previous read-only flag on success/failure, isolate Prod, and serialize against flag changes.
Material identifiers	Within each tenant *dataset generation*, assign sequential immutable eight-digit zero-padded material IDs beginning 00000001; never reuse inside that generation. Prod never resets; a full Test reset intentionally restarts *all* Test-local business ID sequences from 00000001. On XLS import: blank material ID creates; existing ID updates; unknown explicit ID yields a row-level error.
Master-data XLS workflow	Download documented template, edit, upload, run mandatory full dry run. Report all sheet/row/field errors, duplicate business keys, missing references/import dependencies, and create/update/unchanged previews. All rows must pass; then one atomic upsert, rejecting if referenced/affected records changed since dry run. Other master IDs are unique within tenant.
BOM workbook	Two sheets: logical BOM headers and complete component lines. Blank BOM ID creates; existing ID identifies a new revision on actual change. Temporary Import Reference links new header and components until permanent ID exists. Omitted components are removed from new version. Report added/changed/removed lines in dry run. No revision when imported BOM is identical to current one.
Stock XLS, opening balances and corrections	Users/admins can upload **desired factual on-hand stock**, not adjustment deltas, for selected material/location positions including partial locations. System calculates adjustments, requires reason, and posts audited inventory movements; missing positions are untouched. Same complete dry-run/atomic/optimistic checks. No ordinary in-transit correction.
UI and preferences	Two primary maintenance screens (Master Data for admin; Transactions for all) plus inventory report, latest-MRP report and audit viewer. Daily data with weekly viewing option: **sum daily flows** (orders, forecast, receipts, issues); show last-day projected balance and the minimum daily balance/peak

	shortage reached during the week. Never sum daily balances or seven duplicate snapshots of a persistent shortage. Per-user English/Polish (English default), dark/light mode.
In-app documentation	Full searchable bilingual help from every screen, guides, examples, contextual tooltips and descriptive errors. Cover standard transactions, dry runs, corrections including temporary positive-stock adjustment followed by full reversal and reversal of that temporary adjustment, Test reset generations, provisional vs executable MRP, UI-only token creation and newcomer onboarding.
Exact footer on every application page	Caveman Planning Good! Made in Jozefoslaw with Tokens, Coffee and OpenMercato Skills during AITechLeaders training. Include login, admin, reports, help and maintenance views, and retain hyperlinks on the named phrases.

06 / API, audit, integrity and idempotency

Area	Required MVP behavior
Complete UI/API parity	Every business/admin operation, reports, audit reads, help, MRP, reversals, spreadsheet dry-run/import and token *listing/revocation* has a bearer-token API equivalent with identical authorization. **Sole exception: token creation is UI-only.** No bearer-authenticated API endpoint may mint a new token; issuance needs an explicitly confirmed action in the authenticated interactive UI session. API provides no extra privilege, with documented stable errors and no silent partial success.
Self-service API tokens	Active user/admin creates any number of unique-per-user named bearer tokens **only from their authenticated interactive UI session, with explicit confirmation**, never through bearer-token API. Token listing/revocation is available in UI and bearer API. Never-expiring tokens, secrets shown once, last-used tracked; inherit user permissions and both-tenant availability. Audits identify account and (for API changes) token ID/name, never secret. UI guards duplicate submissions.
Idempotency on all API writes	Every *available* mutating bearer API requires a scoped idempotency key, including imports, stock posting, order creation, automatic MRP, tenant flag change, token revocation and admin operations; bearer token creation is unavailable. Scope to account + tenant and, for Test, dataset generation (Global for global actions). Save request identity/result for **24 hours**; identical replay within same generation returns original result; different action/payload conflicts. After reset, old Test-generation requests fail generation precondition

	even if an ID repeats; do not silently execute after clearing expired records. UI guards duplicate submission.
Optimistic edits and freshness	UI and API use record versions; stale changes rejected (API HTTP 409). Successful XLS dry runs cannot commit after related record changes or Test dataset regeneration. MRP proposed orders revalidate current stock/demand/documents, rounded quantities and tenant generation; stale or provisional recommendations are not creatable.
Atomic business operations	Production completion, multi-line GR/GI, inventory/transfer movement, reversal, XLS import and auto-MRP order batch update linked documents, inventory ledger and audit **atomically** or not at all. Physical and in-transit inventory can never be negative.
Tenant audit	Append-only snapshot change events, including creation/tombstones, before/after, tenant, timestamp, account/session, UI/API origin and token ID/name when API. Separate from business relational records and inventory movement ledger. Every user can read tenant audit, never edit. Prod retained indefinitely; Test cleared on reset.
Global audit	One Global scope for account creation/deactivation/password resets, token create/revoke, tenant read-only toggles and Test resets. Indefinite retention even after reset, visible read-only to both roles. Never store password or token secrets or credential hashes in audit.
Shared account attribution	Track user account, session and API token, not the human identity of someone sharing credentials. Deactivation/password reset invalidates all session/token credentials instantly.
Read-only gate and global exceptions	Read-only enforces no tenant data modifications even for admin, through UI/API. Existing report and audit reads allowed; admin global account actions and toggle remain usable. Reset temporarily overrides Test data writes internally with exclusive operation lock.
Reference and history integrity	All business references remain tenant- and Test-generation-consistent. Existing documents may complete or reverse with inactive masters; new ones cannot select them. A material with only an active BOM reference may be deactivated if there is no stock/in-transit or unfinished *direct* operational document blocking it. The affected recipe is ineligible for new MRP Make selection; use an eligible alternative or raise Inactive BOM Component / Missing Usable BOM. If an *existing pinned production order* still references that now-inactive component, retain the order but flag its output At-Risk Existing Production, do not treat it as dependable supply or auto-create duplicate replacement. Historical revisions/postings never rewrite; reversals append.

07 / Reporting and nonfunctional requirements

Area	Decided behavior
Current stock report	Show today's physical on-hand by material/location; show in-transit amounts separately. Provide access to signed stock-movement ledger and source document references.
Last MRP report	Daily 90-day projection all sites, filters and weekly display (sum flows, week-end stock and minimum/peak deficit; never sum stock balances). Firm/forecast/safety demand, tentative/dependable/at-risk existing production supply, separately labeled hypothetical multilevel plans; shortages and pre/post recommendation balances; lot-rounded BOM alternatives, inactive-input rejections, blocked-at-risk vs executable actions and overdue/late/beyond-horizon dates.
Production variance reporting	Compare recorded production order target and pinned BOM expected consumption to actual output/input, including extra components and zero-output waste. No finance/capacity. The existing full-once reversal rules remain; rare corrections may use temporary reasoned positive-stock adjustments which must subsequently be reversed.
Audit report	Read-only for all users; show tenant-scoped and Global events with timestamp, account, session, UI/API/token attribution and before/after values. No password or token secret disclosure.
Performance and data volumes	Best effort for small-enterprise use with sensible performance practices. No numeric targets or contractual SLA.
Navigation, help, language, theme	Contextual tenant indicator; searchable bilingual Help/tooltips and guided examples; per-user Polish/English (English default); selectable dark/light modes. Host platform implements framework and design system.
Footer	Caveman Planning Good! Made in Jozefoslaw with Tokens, Coffee and OpenMercato Skills during AITechLeaders training — exact sentence appears on all app pages, with working outbound links.

08 / MVP gap review — decisions closed

Outcome: The previously identified MVP open questions have been resolved, by explicit product-owner answers or the small operational defaults below. This is not an invitation to add capabilities. Full business rules are in sections 02–07 and the decision register and acceptance scenarios.

Agreed during grilling

Area	Decided behavior
BOM	Per output BUoM; immutable new revision on real edit only; same-priority overlap prohibited; circular recipes rejected; 6-decimal exact quantities; BUoM converted at entry; manual BOM choice possible.
MRP	90-day daily, all tenant locations, rerun nets real documents, forecast consumption with expired past residual and overdue firm demand, material-site parameters, demand priority, lot-rounded BOM feasibility, complete provisional multilevel plan separate from executable supply, start-day BOM inputs, overdue/late/beyond-horizon warnings, tentative PR and dependable PO, atomic eligible auto-creation and stale checks.
Inventory and corrections	Physical never negative (MRP projection may be), today-only postings, GR/GI rather than shipment changes stock, full-once line/posting reversals, rare audited temporary-stock workaround reversed afterward; transfers with partial dispatch/receipt and in-transit, loss means Closed Early; inventory XLS uses desired factual on-hand and atomic dry run.
Orders and parties	Minimal four statuses and quantity-based partial progress; multi-line customer/PO, single-location header; direct PO, flexible supplier; over-receipt allowed, outbound over-delivery prohibited; per-line issues/receipts/reversals; no customer return module.
Tenants and security	Two isolated tenants, one admin, shared sessions and unlimited trusted-account tokens; token creation via explicit UI confirmation only and revocation UI/API. Password reset/deactivation invalidates sessions/tokens, admin tenant read-only and atomic fresh Test reset with restarted IDs/generation precondition, Global audit retained, all exposed mutating bearer APIs idempotent for 24h.
Experience	English/Polish, selectable dark/light, complete searchable contextual help, strict branded footer on every application page; no numeric performance SLA.

Minimal implementation defaults chosen for closure

Area	Decided behavior
Business	Use a single, consistent configured application business date to determine

date	'today' for UI and API. Daily buckets only; no intra-day inventory planning. Post physical transactions today only, never backdate or future-date.
Precision and determinism	Store exact six-decimal BUoM quantities; alternate-UoM input converts via decimal full precision, half-up once on entry. Round aggregated calculated component needs and supply *up*, then enforce minimum/lot multiple on final output before evaluating full BOM feasibility. Never use float arithmetic or revalue posted history. Stable-ID tie-breakers do not change equal same-day business priority.
MRP pipeline	Tentative PR prevents duplicate PRs but does not make components dependable; only physical stock and already-scheduled dependable receipts/issues can satisfy executable BOM feasibility. Show full hypothetical multilevel plan separately without netting its output; no parent becomes executable from same-run hypotheticals. Stale/out-of-generation suggestions fail.
Posting integrity	Full-once posted-line/production-completion reversal; mandatory reason and compensating postings. Temporary positive-stock adjustment may permit reversal when historical output was partly used but must itself be reversed after corrected posting; all balances stay nonnegative. Statuses recalculate; transfer write-off or abandonment resolves as Closed Early, not Completed.
Imports and concurrency	Reject duplicate master keys in workbook; allow intended BOM header/line references; full row errors and change previews; source version and Test dataset-generation guard for every dry run/commit. Identical BOM imports are No Change.
No added modules	No finance, returns/refunds, approval workflows, warehouse bins/batches, reservations, machine capacity, supplier pricing, SSO, automatic transfers, multi-admin, or tech/deployment design.

> **Readiness:** Functional MVP baseline is closed. An implementation may choose ordinary internal technical details consistent with these rules, but should not invent additional business requirements or broaden scope. The companion decision register records status and acceptance tests.

Provenance and companion documents

Compiled from the CPG product-owner requirements interview and extensive follow-up grilling, including the product-owner's instruction to close remaining MVP gaps without scope creep. This document is the functional and nonfunctional baseline; any underlying framework, programming languages or deployment model remain the platform's responsibility.

Detailed visual conceptual model and entity relationships →

Decision register, resolved defaults and 30 compact acceptance examples →

Comprehensive acceptance catalogue — 535 Given / When / Then scenarios across 19 functional areas →

Caveman Planning Good! Made in Jozefoslaw with Tokens, Coffee and OpenMercato Skills during AITechLeaders training